

Leveraging artificial intelligence in library systems: insights from ICTIS

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1. Overview ICTIS

The 10th International Conference on ICT for Intelligent Systems (ICTIS 2025) marked a decade of this annual forum uniting ICT and intelligent systems research. In 2025, the conference adopted a “hybrid format” – a physical event held on 23 May at Cornell University’s NYC-ILR Center and a two-day virtual session (23–24 May) to maximize global reach. This bi-phasic approach (Bangkok in April and New York in May) reflects ICTIS’s evolution into a truly international platform. By bringing together researchers, educators and practitioners worldwide, the conference has continually expanded its impact. All accepted contributions are published in a Springer volume (SIST series) and indexed by major databases (SCOPUS, EI, INSPEC, etc.), ensuring enduring visibility of ICTIS scholarship. In effect, the 10th edition underscores ICTIS’s long-term role as a vehicle for disseminating cutting-edge ICT innovations, with particular benefits for library and information science (LIS) communities seeking to adopt intelligent technologies.

The hybrid model proved successful in fostering “international collaboration.” Attendees hailed from at least seven continents (Asia, Europe, North America, etc.), with participants reported from India, Saudi Arabia, Japan, Korea, USA, UK, Uzbekistan and beyond. University researchers (e.g. from Serbia, USA, India) mingled with industry and publishing professionals (e.g. Springer editors) and government lab experts. For example, the keynote roster alone spanned academia and industry: Prof. Milan Tuba (Singidunum University, Serbia), Dr Ana Trišović (MIT, USA) and Dr Eva Tuba (Trinity University, USA) represented universities, while Dr Pravir Malik (QIQuantum, USA) and Aninda Bose (Springer Nature, UK)

brought industry and publishing perspectives. Such diversity – by country, institution and role – created rich networking opportunities. Attendees reported establishing contacts across borders and disciplines, which is expected to seed future joint projects. In summary, ICTIS 2025 was built on the conference’s decade-long legacy by combining a broad, international audience with high-quality outputs (Springer proceedings) to promote global knowledge exchange in intelligent systems.

2. Keynote sessions

Dr Ana Trišović (MIT, USA) – Trišović’s keynote explored the intersection of AI, scientific research and open data. She emphasized that modern AI tools must be developed within *open and reproducible frameworks* to benefit science and society. Quoting from her public profile, she noted that her work “focuses on exploring the intersection of artificial intelligence (AI), science, and society, aiming to advocate for improved research practices.” In particular, Trišović highlighted concerns about AI-driven “de-democratization” of knowledge and underscored the *importance of open-source solutions*. She urged information professionals to consider how AI can both empower and potentially exclude users if proprietary algorithms dominate. In library systems, her message reinforces the need for transparent AI services and open data policies, so that innovations in cataloging or analytics remain accessible and reliable.

Dr Pravir Malik (QIQuantum, USA) – Malik presented a visionary talk on quantum computing and “Quantum Intelligence.” Drawing on his decade of work, he described an avant-garde quantum architecture that he calls the *Quaternary Interpretation of Quantum Mechanics* (QIQM). Unlike conventional

binary or qubit systems, this approach uses a “fourfold logic” (with quaternary gates) embodied in a “reconfigurable atom-denominated quantum computer (RADQC).” He asserted that such a system could *revolutionize* computational power and AI. For instance, Malik envisioned the RADQC “will give rise to a vast array of unforeseen technologies, processes, and devices in areas of computation, genetics, artificial intelligence, and transhumanism.” In concrete terms, he suggested applications ranging from accelerated drug design to novel material discovery through advanced AI models running on quantum hardware. While these ideas are at the research frontier, Malik’s keynote underscored a critical insight for librarians and information scientists: the exponential growth in compute (via quantum) may one day enable library systems to analyze massive data sets or complex knowledge graphs far beyond today’s capabilities. His talk was peppered with enthusiastic detail (e.g. discussing a pending patent for the RADQC) and left the audience reflecting on long-term “what if” scenarios for AI (Table 1).

Prof Milan Tuba (Singidunum University, Serbia) – Tuba’s keynote, titled “Machine Learning for Intelligence Systems with Digital Images,” focused on the role of machine learning (ML) in computer vision and image analysis within intelligent systems. He reviewed classic and modern methods: early image processing relied on hand-crafted feature extraction (edges, textures, shapes) as an intermediary step in recognition. As one conference panelist observed (drawing on Dr Eva Tuba’s earlier work), “we find edges, texture, color contrast [...] and then use machine learning to learn the pattern.” Tuba noted that this manual feature engineering has largely been supplanted by deep learning: “recent models such as convolutional neural networks automatically extract features

Table 1
Keynote speakers and topics at ICTIS 2025

Speaker	Affiliation	Keynote focus
Dr Ana Trišović	MIT (FutureTech Lab), USA	AI, science and society; open-source and reproducibility
Dr Pravir Malik	Founder, QIQuantum, USA	Quantum computing architectures (quaternary logic); AI's future
Prof Milan Tuba	Singidunum University, Serbia	ML for computer vision; deep learning for images
Dr Eva Tuba	Trinity University, USA	AI model optimization; adversarial robustness; content protection
Mr Aninda Bose	Executive Editor, Springer Nature, UK	Ethics of research publishing; global collaboration

Source(s): Author's own work

[...] and significantly better accuracy is achieved.” He illustrated this shift with examples in biomedical imaging (tumor segmentation using magnetic resonance imaging) and optical character recognition for digitizing texts. In discussing these techniques, Tuba connected to library systems by highlighting how similar ML approaches can improve digital archives: e.g. automatically tagging images and scanned documents, enhancing search in digital collections or enabling AI-driven optical character recognition for historical books. Throughout, he stressed that the “intelligence” in intelligent systems often derives from such ML pipelines, especially for multimedia information in libraries (Table 1).

Dr Eva Tuba (Trinity University, USA) – Dr Eva Tuba’s keynote addressed cutting-edge issues in AI model development and data security. She drew on her award-winning research to describe methods for optimizing and hardening AI systems. For example, she detailed work on AI *model tuning*, using evolutionary or swarm optimization algorithms to find “lightweight models that reduce computational demands while preserving quality.” Her group has also focused on transfer learning and domain adaptation to improve model reliability under different data conditions. Another major theme was adversarial robustness: strengthening models “against vulnerabilities” so that they maintain accuracy under manipulation. Crucially, for the library context, Eva Tuba also presented innovative *content-protection* techniques. One discussed project uses AI “data poisoning” to safeguard copyrighted images: by subtly distorting original art before public posting, it thwarts unauthorized AI art generators. Another uses diffusion-model-based embeddings to watermark or trace digital content. Her forward-looking closing point was on explainable AI, emphasizing that

librarians and technologists must eventually demand transparency in any AI tools they deploy. Overall, her keynote highlighted the dual needs of *efficiency and trustworthiness* in AI systems – lessons directly relevant to libraries automating metadata generation or surveillance detection, where model performance must be balanced by integrity and explainability (Table 1).

Mr Aninda Bose (Springer Nature, UK) – Bose gave a keynote on “Ethics in Research Publishing”, reflecting his role as a publishing executive. He reminded the audience that behind all technological advances lies a scholarly infrastructure that must uphold integrity. Bose discussed issues like authorship ethics, plagiarism and the responsible use of generative AI in writing. (His LinkedIn post modestly states only that it was “good to see a great gathering of authors from different parts of the world,” but conference attendees reported that he issued a strong call for transparency and collaboration among authors, publishers and libraries.) In remarks to the press, he linked ethical publishing to industry trends such as open access and data sharing, noting that libraries serve as custodians of knowledge. The takeaway for librarians is clear: as AI-driven tools (for peer review or reference checking) emerge, the community must jointly enforce ethics. Bose’s emphasis on “publishing ethics” resonates with libraries’ stewardship role in guiding researchers through the new AI-enhanced scholarly landscape (Table 1).

3. Thematic sessions and workshops

Beyond the plenaries, ICTIS 2025 featured *technical sessions and workshops* spanning a wide range of topics on intelligent systems. Session presentations aligned with the speakers’ interests and the conference scope (Plate 1). In particular,

ML and *AI* dominated many tracks. Attendees learned about new ML architectures for pattern recognition, deep learning for big data analytics and intelligent agents for semantic search. For example, papers showcased neural network classifiers for document image processing and predictive analytics in sensor networks. One participant’s summary captured the breadth: presentations covered “sensors, IoT, machine learning, deep learning, web security, and information management systems.” Indeed, workshops on *Internet of Things (IoT) and sensor networks* were prominent, demonstrating how distributed devices can feed data into intelligent systems (e.g. smart campus or smart library prototypes with environmental sensing).

Other sessions highlighted *data and information management*. These included big data analytics, cloud-computing frameworks and knowledge management techniques. Several papers described case studies of large-scale data curation: for instance, using Hadoop clusters for library catalog data or graph databases for linking bibliographic records. Sessions on the *Semantic Web and linked data* explored how ontologies and AI can improve metadata interoperability. In parallel, cybersecurity and privacy were major themes: workshops introduced blockchain-based solutions for trust and integrity in data sharing, and panels addressed ethical AI, consistent with broader concerns. As one finds in LIS literature, interdisciplinary collaboration was a focus: presentations often tied ICT research back to library needs.

Digital transformation was a cross-cutting topic. Sessions on *library technology and services* emphasized how ICT innovations can modernize libraries. For example, speakers demonstrated deploying AI chatbots for user inquiries and virtual assistants for reference services. One workshop illustrated building a cloud-based digital repository

Plate 1. Group photo with invited speakers of ICTIS, 2025, USA



Source: Author's own work

for a university library, streamlining archival workflows. Notably, case studies presented at the conference were highly relevant to LIS. A librarianship perspective emerged in talks on automated content tagging, advanced search algorithms and AI-driven user personalization. For instance, an industry-academic team reported applying convolutional neural networks to automatically classify and enhance historical newspaper archives. Librarians would recognize this as akin to modern digital preservation tools. Likewise, a study on securing institutional archives with IoT monitoring (temperature/humidity sensors) linked smart-environment IoT concepts to preservation. Throughout the sessions, the idea that “AI in data curation and discovery” is crucial for libraries was echoed. In summary, the thematic program showcased that *intelligent system technologies* are advancing rapidly in areas directly applicable to libraries: data/information management, digital

infrastructures and service automation, all illustrated with practical examples and case studies.

4. Global participation and networking

ICTIS 2025 drew a *diverse global* audience (Plate 2). Presentations and attendee reports indicate participants from at least Asia (India, Saudi Arabia, Japan, Korea, Uzbekistan), Europe (Serbia, UK, etc.), North America and beyond. These researchers represented a mix of institutions: universities (e.g. Cornell, MIT, Trinity, Singidunum), research institutes (e.g. FutureTech Lab at MIT, QIQuantum Labs) and industry (Springer Nature, tech startups). Professional roles ranged widely – faculty professors and PhD students, research scientists, as well as industry technologists and editors. For example, the keynote roster included both academic and practitioner leaders: Prof Milan Tuba of Singidunum University (an

academic project head) and Dr Ana Trišović of MIT on the academic side; contrasted with Dr Pravir Malik (QIQuantum founder) and Mr Aninda Bose (Springer publishing editor) on the industry side. Librarians and information professionals were also present, leveraging the hybrid format to attend sessions remotely.

This mix facilitated networking across regions and sectors. Conference social events, plenary Q&As and digital forums enabled participants to exchange contact information and discuss joint interests. Early indications are that several collaborative initiatives emerged: for instance, a multi-institution proposal on AI-driven library analytics is reportedly in discussion, and an editorial partnership between ICTIS organizers and LIS journals is under consideration. Attendees commented on LinkedIn about forming new connections (e.g. between US and Indian universities). In summary, ICTIS 2025 served as a nexus for interdisciplinary

Plate 2. Group photo of ICTIS 2025, USA



Source: Author's own work

networking, breaking down geographic and sectoral barriers and planting seeds for future international projects in intelligent systems and library technology.

5. Publication and proceedings

In line with past ICTIS events, *all accepted papers* of ICTIS 2025 will appear in a Springer conference volume (Lecture Notes in Intelligent Systems series – Springer's Smart Innovation, Systems and Technologies). The organizers explicitly note that “all ICTIS 2025 presented papers will be published in conference proceedings by Springer SIST.” These proceedings are already scheduled to be indexed in major scientific databases (SCOPUS, EI Compendex, INSPEC, etc.), ensuring broad discoverability. For LIS professionals, this means that the conference content will be easily accessible via university library subscriptions and online academic

search tools. The availability of the Springer eBook and hardcover editions also means that institutional libraries can acquire the volume for their collections. The Springer imprimatur further signals rigorous peer review and quality, making the proceedings a valuable resource.

Moreover, by publishing in an indexed venue, ICTIS 2025 papers contribute to the LIS literature on emerging technologies. Librarians can cite these papers in their own research, and information science courses can incorporate findings from the conference. In practical terms, the open Springer volume (with ISBN and ISSN) and its indexing allow any scholar or librarian to retrieve specific papers on AI, IoT or other relevant topics covered at ICTIS 2025. This amplifies the event's impact: rather than remaining an isolated event, the research presented becomes part of the continuous scholarly record. In summary, the Springer proceedings

provide enduring value and accessibility to both academics and practicing librarians interested in the intersection of AI and library systems.

6. Relevance to library and information science

ICTIS 2025's themes dovetail closely with current trends and challenges in LIS. The emphasis on AI, data-driven systems and digital infrastructure mirrors the LIS community's agenda for modernization. For example, the conference's focus on AI *and data analytics* speaks directly to how libraries are beginning to apply AI for cataloging, discovery and user services. A LibrarIN report observes that AI is already “transforming library operations” by enabling “automated cataloguing, content tagging, and the digitisation of historical texts” – capabilities that were discussed at ICTIS in various technical talks. Similarly, emerging topics like IoT have clear implications for “smart libraries”

(automated climate control, RFID tracking, etc.), aligning with ICTIS sessions on sensor networks. In short, the innovations highlighted – from intelligent search algorithms to networked smart devices – map onto the digital transformation trajectory of libraries.

Conferences like ICTIS also resonate with broader industry data. A 2024 Clarivate “Pulse of the Library” survey found that over 60% of libraries are actively evaluating or planning AI integration. Librarians rank AI as a top tech priority: key goals are supporting student learning (52%), research excellence (47%) and content discoverability (45%). These priorities parallel the goals of ICTIS presentations on learning analytics, research tools and enhanced search. The conference’s interdisciplinary mix reflects the survey’s finding that collaboration between IT departments and library leadership is crucial for AI adoption. Likewise, concerns raised in the library community (skills gaps, limited budgets) were echoed at ICTIS in discussion forums. Clarivate’s report highlights that nearly half of libraries cite “shrinking budgets” and a lack of AI training as major challenges. ICTIS speakers likewise emphasized the importance of education and ethical use as AI expands. As Derek Brown of Rochester Hills Public Library noted: “The future role of the librarian will be characterised by [...] new, technology-driven tasks [...]. Even with AI, librarians will continue to be indispensable guides.” This insight – that human expertise remains central – was reflected in the conference’s balanced portrayal of AI as a tool (not a replacement) for information professionals.

Practically, LIS professionals can draw several implications from ICTIS 2025. The demonstrated AI techniques suggest new services: for instance, using NLP and ML to enrich metadata, recommending related resources automatically or using computer vision to enhance digital exhibits. The strong focus on data management technologies advises librarians to invest in scalable infrastructures (cloud services, linked data platforms) to support such tools.

Presentations on cybersecurity and ethics remind libraries to safeguard patron privacy as they adopt new systems. And the attention to open-source and reproducibility (from Trišović’s session) underscores the value of collaborative development, a principle that could guide libraries developing or selecting AI tools. In essence, ICTIS 2025 not only showcased what AI and intelligent systems can do, but also highlighted *how librarians and information managers must evolve* – updating skills, infrastructures and policies – to harness these technologies. The overlap between ICTIS’s cutting-edge topics and the LIS sector’s digital transformation agenda confirms that the conference’s insights are highly relevant and actionable for today’s libraries.

7. Conclusion

The 10th International Conference on ICT for Intelligent Systems (ICTIS, 2025) demonstrated a remarkable confluence of technological innovation, interdisciplinary collaboration and global engagement, with a specific emphasis on the integration of AI in library and information systems. Through keynotes, workshops and technical sessions, the event highlighted transformative advancements in AI, ML, quantum intelligence, content protection and ethical publishing – all of which bear direct relevance to modern library services. The diversity of participants – spanning academia, industry and publishing – amplified the conference’s impact, fostering collaborations that are expected to influence future research and practice in intelligent information systems.

For the LIS community, ICTIS 2025 offered vital insights into leveraging intelligent technologies to enhance metadata curation, digital preservation, user personalization and ethical AI integration. The emphasis on open-source tools, explainable AI and smart infrastructure aligns seamlessly with ongoing efforts to modernize libraries into dynamic, user-centric digital ecosystems. Moreover, the publication of conference proceedings in

indexed Springer volumes ensures sustained access and scholarly contribution, bridging theory with practical applications.

In essence, ICTIS 2025 not only celebrated a decade of progress but also charted a forward-looking roadmap for integrating AI in library contexts. It reaffirmed the librarian’s evolving role as both a technologist and a steward of information ethics. As AI and intelligent systems continue to shape knowledge discovery, events like ICTIS will remain critical in guiding LIS professionals through the ethical and technical terrain of digital transformation.

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